

HMG-CoA reductase inhibitors and the risk of hip fractures in elderly patients.

SCIFACT-EVIDENCE-4687948

CONTEXT Recent animal studies have found that 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) lipid-lowering drugs (statins) substantially increase bone formation, but whether statin use in humans results in clinically meaningful bone formation or a reduction in the risk of osteoporotic fractures is not known. **OBJECTIVE** To determine whether the use of statins is associated with reduced hip fracture risk. **DESIGN** Case-control study. **SETTING AND PATIENTS** A total of 6110 New Jersey residents aged 65 years or older and enrolled in Medicare and either Medicaid or the Pharmacy Assistance for the Aged and Disabled program. Case patients (n=1222) underwent surgical repair of a hip fracture in 1994. Control patients (n=4888) were identified at a ratio of 4:1 and frequency-matched to case patients for age and sex. **MAIN OUTCOME MEASURE** Adjusted odds ratio (OR) of hip fracture by statin use in the 180 days and 3 years prior to the index date (the earliest date of admission for surgery), adjusted for demographic and clinical characteristics and health care utilization. **RESULTS** Use of statins in either the prior 180 days (adjusted OR, 0.50; 95% confidence interval [CI], 0.33-0.76) or prior 3 years (adjusted OR, 0.57; 95% CI, 0.40-0.82) was associated with a significant reduction in the risk of hip fracture, even after controlling for variables such as race, insurance status, psychoactive medications, estrogen and thiazide use, ischemic heart disease, cancer, and diabetes mellitus. No significant relationship was observed between use of nonstatin lipid-lowering agents and hip fracture risk. Clear relationships were observed between the degree of reduction in hip fracture risk and the extent of statin use; there was no evidence of such relationships with nonstatin lipid-lowering agents. After adjusting for extent of statin use in the prior 3 years, current use (on the index date) was associated with a 71% reduction in risk (adjusted OR, 0.29; 95% CI, 0.10-0.81). The relationship between statin use and hip fracture risk persisted after controlling for variables such as the number of medications, the Charlson comorbidity index score, and hospitalization or nursing home stay in the last 180 days, as well as after excluding patients who were in a nursing home prior to their index date or who died in the year after their index date. Use of nonstatin lipid-lowering agents was not observed to be associated with reduction in hip fracture risk in any of these alternative models or analyses. **CONCLUSIONS** These findings support an association between statin use by elderly patients and reduction in the risk of hip fracture. Controlled trials are needed to exclude the possibility of unmeasured confounders. JAMA. 2000;283:3211-3216